



SCIENCEDOJO PRODUCT DOCUMENTATION

Lesson Plan Builder

How curriculum-aware client intake becomes an accurate, reviewable lesson plan

Developer guide

Version 1.0 | 21 September 2026

IMPLEMENTATION STATUS

Curriculum-aware intake implemented

Next component: lesson-plan generation, validation, and tutor review

SECTION 01

Purpose and product boundary

This document explains how the existing intake implementation should feed a separate automatic lesson-plan builder. It is the handoff contract between booking intake and lesson generation.

Core rule

The generator must consume normalized, server-validated fields. It must not reverse-engineer curriculum, exam board, tier, topic, or learner baseline from the parent's free-text message.

The intake feature is already responsible for collecting and storing the educational facts needed to ground a lesson. The lesson-plan builder is responsible for turning a complete snapshot into a structured plan, validating that output, and presenting it to the assigned tutor for review.

In scope for the builder

- Read the immutable learning-context snapshot stored on a booking.
- Resolve curriculum and specification context from stable keys.
- Use uploaded materials as private supporting evidence.
- Generate a structured lesson plan that fits the booked duration and learner baseline.
- Explain which intake fields and source materials influenced the plan.
- Require tutor review before the plan is treated as teaching-ready.

Out of scope

- Inferring missing curriculum details with confidence when the intake is marked `needs_clarification`.
- Replacing tutor judgement, safeguarding decisions, assessment diagnosis, or special educational needs assessment.
- Publishing or exposing private homework, teacher feedback, assessment papers, or learner data outside the booking participants.

Definition of done

A generator can produce a useful draft without parsing UI labels or free text, and a tutor can see the assumptions, sources, objectives, sequence, checks for understanding, and adaptations before accepting the plan.

SECTION 02

Current intake architecture

One education taxonomy and one controlled selector now serve tutor booking, free assessment, and Practice Dojo.

1

Learner identity and baseline

Learner profile supplies name, school year or grade, pathway defaults, support preferences, and accommodations.

2

Curriculum hierarchy

Client selects curriculum pathway, stage or qualification, exam board where applicable, subject, subject route, tier, topic, and optional specification code.

3

Lesson intent

Client supplies purpose, desired outcome, confidence, exact difficulty, attainment, assessment context, and longer-term goals.

4

Server validation

Server validates the combination independently of the browser and normalizes it to schemaVersion 2.

5

Immutable storage

The booking receives an immutable learning_context JSON snapshot. Free-assessment leads store the same education selection without creating an account.

6

Tutor visibility

The assigned tutor sees the route, subject, topic, goal, confidence, missing fields, and private materials before accepting.

Implemented surfaces

Surface	Purpose	Implementation
Tutor booking	Authenticated lesson request plus reusable learner profile	components/CalendlyBookingWizard.tsx
Free assessment	Accountless lead capture using the same education hierarchy	app/free-assessment/FreeAssessmentForm.tsx
Practice Dojo	Curriculum-aligned practice selection	app/ai-practice-studio/QuestionGenerator.tsx
Shared selector	Controlled dependent dropdowns and clarification messaging	components/EducationPathwayFields.tsx
Validation	Taxonomy, normalization, compatible combinations	lib/educationTaxonomy.ts
Lesson context	Booking-level contract and conditional intake rules	lib/lesson-request-intake.ts
Database	Profiles, snapshots, specification registry, private materials	sql/051_lesson_request_intake.sql

SECTION 03

Education hierarchy and terminology

The field order prevents the common error of treating an exam board, qualification, curriculum, and tier as the same thing.

Field	Meaning	Examples
Curriculum pathway	Education system or international programme family	England, Wales, Northern Ireland, Scotland, Cambridge International, Pearson International, IB
Stage or qualification	The learner's programme or assessed qualification	KS2, KS3, GCSE, A Level, National 5, IGCSE, IB DP
Exam board / awarding body	Organisation owning the specification and assessment	AQA, Pearson Edexcel, OCR, Eduqas, WJEC, CCEA
Subject	Broad academic subject	Mathematics, Physics, English, Geography
Subject route	A distinct course within a subject family	Combined Science, English Literature, Further Mathematics, IB Mathematics AA
Tier / course level	Course depth or assessment tier when applicable	Foundation, Higher, Core, Extended, SL, HL
Specification	Versioned syllabus used for retrieval and provenance	A specification code supplied by the client or a resolved active version ID
Topic / subtopic	Immediate content focus	Algebra / quadratic factorisation

Why exam board is conditional

KS1, KS2, and KS3 follow a curriculum rather than an exam-board specification. Exam board becomes mandatory only for examined routes where the provider affects content, terminology, paper structure, or required practicals.

Supported pathway families

- England National Curriculum: KS1, KS2, KS3, GCSE, AS Level, A Level.
- Curriculum for Wales: Primary, Secondary, GCSE, AS Level, A Level.
- Northern Ireland Curriculum: Foundation, KS1 to KS4, GCSE, AS Level, A Level.
- Scotland Curriculum for Excellence: Primary, Broad General Education, National 4, National 5, Higher, Advanced Higher.
- Cambridge International: Early Years, Primary Stages 1 to 6, Lower Secondary Stages 7 to 9, IGCSE, O Level, International AS and A Level.
- Pearson Edexcel International: iPrimary, iLowerSecondary, International GCSE, International AS and A Level.
- International Baccalaureate: PYP, MYP, DP, CP.

SECTION 04

Normalized input contract

The generator should depend on stable keys and schema version, never display labels or raw FormData.

```
type EducationSelectionSnapshot = {
  schemaVersion: 2;
  curriculumKey: string | null;
  stage: string | null;
  awardingBodyKey: string | null;
  level: string | null;
  subject: string;
  subjectVariant?: string;
  specificationVersionId: string | null;
  specificationCode?: string;
  topic?: string;
  subtopic?: string;
  intakeStatus: "complete" | "needs_clarification";
  missingFields: string[];
};
```

The booking's LessonRequestLearningContext adds learner and lesson fields

- learnerId, learnerName, schoolYear
- lessonPurpose, lessonGoal, confidence, difficultyDetails
- currentAttainment, targetAttainment, assessmentDate, assessmentDetails
- homeworkInstructions, recentScore, teacherFeedback, lostMarksOn
- longerTermGoal, priorityTopics, importantDeadline
- supportPreferences, accommodations, previousApproaches, additionalContext
- materialIds, intakeStatus, missingFields

Compatibility behavior

- Existing booking description remains available and is copied to additionalContext.
- Legacy assessment-lead curriculum and subject columns remain human-readable summaries.
- New generator code should check schemaVersion before reading v2-specific properties.
- Existing bookings without learning_context should remain visible but are not automatically plan-ready.

Generator entry condition

Only generate automatically when learning_context exists and intakeStatus is complete. A needs_clarification request should create a clarification task, not an invented curriculum assumption.

SECTION 05

How each input improves the lesson plan

Every collected value should have an explicit generation purpose. Unused data should not be collected.

Input	Use in generation	Risk prevented
School year / grade	Controls language, examples, cognitive load, and safeguarding-appropriate presentation.	Age mismatch and unsuitable explanations.
Curriculum pathway	Selects the education system, terminology, progression model, and expected prior knowledge.	Mixing national and international curricula.
Stage / qualification	Sets expected depth, assessment style, and prerequisite range.	Teaching GCSE depth to KS3 or vice versa.
Awarding body	Targets provider-specific specification structure and assessment language.	Using the wrong content coverage or paper conventions.
Subject route	Distinguishes courses that share a broad subject name.	Confusing Language with Literature, or Combined with Separate Science.
Tier / level	Sets ceiling, complexity, and question style.	Higher-tier content in a Foundation session, or SL/HL mismatch.
Specification version	Anchors retrieval to the active syllabus and supports provenance.	Outdated or silently guessed curriculum content.
Topic / subtopic	Defines the content boundary for objectives and activity sequence.	Generic lesson plans.
Lesson purpose	Selects the lesson pattern: teach, catch up, review, revise, assess, or stretch.	A revision lesson structured like first teaching.
Desired outcome	Becomes the measurable success criterion.	Activities without a clear end state.
Confidence	Controls scaffolding, pacing, and amount of guided practice.	Overwhelming or under-challenging the learner.
Difficulty details	Targets the precise misconception or blocked step.	Repeating content the learner already understands.
Current / target attainment	Calibrates challenge and progression steps.	Unrealistic or weak difficulty selection.
Assessment date	Controls urgency, consolidation, and exam-series specification resolution.	Poor pacing before a deadline.
Support preferences	Adapts explanations, checks, visuals, pace, and practice style.	A plan that is technically correct but unusable for the learner.
Materials	Provides the actual question, feedback, worksheet, or syllabus evidence.	Hallucinating what the assignment or feedback says.
Additional context	Adds useful nuance after structured fields are fixed.	Free text overriding validated facts.

SECTION 06

Generation pipeline

Recommended server-side sequence for the lesson-plan builder.

1

Load and authorize

Fetch the booking, confirm requester or assigned tutor access, and reject unrelated users.

2

Check readiness

Require schemaVersion 2 and intakeStatus complete. Return explicit missingFields otherwise.

3

Resolve specification

Use specificationVersionId and optional specificationCode to load the active curriculum record and trusted source references.

4

Retrieve materials

Read only materialIds attached to the booking. Extract text from PDFs/images in a private processing context.

5

Build evidence packet

Combine normalized facts, specification context, material excerpts, booked duration, and lesson purpose. Keep structured fields authoritative.

6

Generate structured JSON

Ask the local LLM for the lesson-plan output contract, not prose-only text.

7

Validate output

Check schema, duration totals, objective alignment, age suitability, source references, and prohibited assumptions.

8

Persist draft and provenance

Save input snapshot ID/hash, model version, prompt version, source IDs, validation result, and generated plan.

9

Tutor review

Allow tutor edit, approve, regenerate with feedback, or request clarification. Record final approval.

Precedence rule

Structured snapshot > verified specification source > uploaded learner material > additionalContext. Never let a free-text note silently replace a validated curriculum key or exam board.

SECTION 07

Recommended lesson-plan output contract

A structured output makes plans editable, testable, and usable by the tutor dashboard.

```
type GeneratedLessonPlan = {
  schemaVersion: 1;
  bookingId: string;
  status: "draft" | "needs_review" | "approved";
  title: string;
  curriculumAlignment: {
    specificationVersionId: string;
    specificationCode?: string;
    objectives: string[];
    sourceIds: string[];
  };
  learnerSummary: {
    startingPoint: string;
    misconceptionFocus: string[];
    adaptations: string[];
  };
  successCriteria: string[];
  requiredMaterials: string[];
  lessonSequence: Array<{
    phase: "diagnostic" | "explain" | "model" | "guided" |
      "independent" | "assess" | "review";
    minutes: number;
    tutorActions: string[];
    learnerActions: string[];
    checksForUnderstanding: string[];
  }>;
  differentiation: {
    support: string[];
    stretch: string[];
  };
  exitCheck: string[];
  followUp?: string[];
  assumptions: string[];
  warnings: string[];
  provenance: {
    intakeSchemaVersion: 2;
    promptVersion: string;
    modelVersion: string;
    generatedAt: string;
  };
};
```

Hard validation rules

- Sum of lessonSequence.minutes must equal the booked duration.
- Every success criterion must be observable during the lesson.
- At least one diagnostic check and one exit check are required.
- The plan must address difficultyDetails directly.
- Any accommodation or support preference must appear as a concrete adaptation, not a label.
- Claims tied to uploaded material or a specification must identify a sourceId.
- Warnings must expose unresolved assumptions; hidden assumptions are validation failures.

SECTION 08

Prompt and retrieval assembly

Keep the prompt modular so curriculum data, learner data, and lesson method can evolve independently.

Recommended prompt blocks

Block	Contents	Authority
System policy	Role, output schema, safety, no-invention rule	Highest
Education route	Pathway, qualification, board, route, tier, specification	Validated snapshot
Learner baseline	Year, confidence, attainment, difficulty, accommodations	Validated snapshot
Lesson request	Purpose, goal, topic, duration, recurrence, deadline	Booking snapshot
Curriculum evidence	Retrieved objectives and specification excerpts	Trusted registry/source
Learner materials	Homework, feedback, assessment excerpts	Private uploaded evidence
Additional context	Parent/student free-text note	Supplementary only
Output instruction	GeneratedLessonPlan JSON schema and validation constraints	Required

Retrieval behavior

- Query curriculum sources using stable keys, not display text.
- Filter by specificationVersionId first; use specificationCode as a stronger discriminator when supplied.
- Chunk source documents by section and retain page/section provenance.
- Retrieve only the selected topic plus prerequisites needed for the learner's stage.
- Do not blend objectives from another awarding body merely because the topic name matches.
- Treat uploaded exam questions as learner-provided material, not as permission to reproduce copyrighted papers broadly.

Local LLM guidance

The model should receive a compact evidence packet with explicit labels and stable keys. Avoid passing the full booking database row or unrelated learner history.

SECTION 09

Clarification and failure modes

The product should fail usefully when the inputs cannot support an exact plan.

Condition	System behavior	User-facing result
intakeStatus = needs_clarification	Do not call the generator	Show missing fields and request clarification
No learning_context	Treat as legacy booking	Tutor can use description; offer structured intake upgrade
Invalid combination	Reject server-side	Explain which selection conflicts
Specification unresolved	Do not guess provider/version	Ask for code, board, or exam year
Material unreadable	Continue only if material is non-essential	Warn tutor and identify failed file
Homework request without useful text/file	Reject as incomplete	Request assignment instructions or upload
LLM output fails schema	Retry once with validation feedback	If still invalid, record failure for tutor
Sequence duration mismatch	Reject generated output	Regenerate with exact remaining minutes
Unsupported subject/specification	Mark clarification or manual mode	Do not label the output curriculum-aligned

Clarification questions should be specific

- Good: 'Is this OCR GCSE Physics, and is the learner studying Higher tier?'
- Good: 'Which question numbers from the homework should the lesson cover?'
- Bad: 'Please provide more details.'
- Bad: asking the client for an internal specification version identifier.

SECTION 10

Privacy, access, and safety

Learner context and uploaded materials are private educational records and must remain scoped to the booking.

Access model

- Requester can view their own learner profiles, booking context, and uploaded materials.
- Only the assigned tutor can view lesson-request materials for that booking.
- Authorized admins can access records for support and safeguarding operations.
- Unrelated tutors and users must receive no metadata, filenames, signed URLs, extracted text, or generated plan content.

Generation controls

- Run authorization before loading any material or generating a plan.
- Use short-lived access to private objects; do not make the storage bucket public.
- Do not place learner names, diagnoses, contact details, or raw files in logs.
- Persist model input hashes and source IDs rather than unnecessary duplicate raw text.
- Allow admins to trace which booking, prompt version, model version, and sources produced a plan.
- Tutor approval is mandatory before a generated draft is shown as final teaching content.

Accommodation wording

Collect teaching-relevant accommodations without requiring a diagnosis. Generate concrete delivery adaptations, but never infer a medical or learning diagnosis from behavior or parent notes.

SECTION 11

Implementation sequence for the generator developer

Build the next component without changing the intake contract.

Phase 1 - Contracts and readiness

- Import LessonRequestLearningContext and EducationSelectionSnapshot from the existing libraries.
- Add a readiness function returning ready, needs_clarification, legacy, or unsupported.
- Define and test GeneratedLessonPlan plus runtime validation.

Phase 2 - Specification and material retrieval

- Create a specification repository keyed by specificationVersionId and optional specificationCode.
- Add private material extraction with source IDs and page/section metadata.
- Build the compact evidence packet and precedence rules.

Phase 3 - Generation and validation

- Version the system prompt and model configuration.
- Generate JSON, validate it, and retry once using validation feedback.
- Store draft, provenance, warnings, validation results, and input snapshot reference.

Phase 4 - Tutor workflow

- Show plan draft next to the original learning brief and materials.
- Support edit, approve, regenerate with feedback, and request clarification.
- Keep an audit trail of generated and tutor-edited versions.

Phase 5 - Controlled release

- Start behind a feature flag with internal/admin and selected tutors.
- Measure readiness rate, clarification rate, generation failure rate, tutor edits, and approval time.
- Do not auto-send plans to parents or students during the initial release.

SECTION 12

Test and acceptance matrix

Tests should demonstrate educational correctness, data integrity, and safe failure behavior.

Area	Acceptance check
Taxonomy	Every pathway exposes only compatible stages, boards, subjects, variants, and tiers.
Non-exam route	KS1 to KS3 submits without an exam board.
Exam route	GCSE/A Level requires or derives a compatible awarding body.
Provider identity	Cambridge International can never validate as OCR.
Unknown values	Not sure produces needs_clarification with explicit missingFields.
Direct action	Server rejects malformed combinations even when UI checks are bypassed.
Specification	Assessment year affects version resolution and ambiguity never silently resolves.
Legacy	Old descriptions and leads still render without a v2 context.
Privacy	Unrelated tutor cannot access material, extracted text, or plan.
Output schema	Invalid phase, missing objective, or malformed provenance is rejected.
Duration	Activity minutes equal the booked lesson duration.
Goal alignment	Success criteria and exit check reflect lessonGoal and difficultyDetails.
Adaptation	Preferences and accommodations become observable teaching actions.
Failure	Generator does not run for needs_clarification or unresolved specification.
Audit	Stored draft records prompt, model, source, and intake schema versions.

Existing verification

The current intake implementation passes TypeScript checks, the automated test suite, targeted lint, a production build, and a public free-assessment smoke test. Generator work should preserve those checks.

SECTION 13

Deployment and handoff checklist

The generator should not be enabled until storage and access controls are in place.

Before deploying the intake schema

- Apply `sql/051_lesson_request_intake.sql` in Supabase.
- Confirm `curriculum_specifications`, `learner_profiles`, `booking_learning_context`, `assessment_lead_learning_context`, and `lesson_request_materials` exist.
- Verify the `lesson-request-materials` bucket is private and RLS policies work for requester, assigned tutor, and admin.
- Verify immutable snapshot triggers do not block unrelated booking or lead status updates.

Before enabling generation

- Backfill or intentionally classify legacy bookings without `learning_context`.
- Populate trusted specification records and source URLs for supported subjects.
- Choose the local model, context limit, prompt version, retry policy, and timeout.
- Create the generated-plan persistence model and access policies.
- Add monitoring for clarification, failures, latency, token use, and tutor edits.
- Complete privacy and safeguarding review for uploaded material extraction.

Developer handoff rule

Do not change the intake contract casually

If the generator needs another field, add it through a versioned contract change, update both forms and server validation, migrate storage, and retain compatibility readers. Do not start parsing `additionalContext` as a substitute.

SECTION 14

References

Authoritative curriculum references and implementation sources used by the intake architecture.

External curriculum references

- Ofqual guide for schools and colleges: <https://www.gov.uk/government/publications/ofqual-guide-for-schools-and-colleges-2026/ofqual-guide-for-schools-and-colleges-2026>
- England National Curriculum: <https://www.gov.uk/national-curriculum>
- Curriculum for Wales: <https://hwb.gov.wales/curriculum-for-wales>
- Northern Ireland statutory curriculum: <https://www.education-ni.gov.uk/articles/statutory-curriculum>
- Qualifications Scotland: <https://qualifications.gov.scot/>
- Cambridge programmes and qualifications: <https://www.cambridgeinternational.org/programmes-and-qualifications/>
- Pearson international qualifications: <https://qualifications.pearson.com/>
- International Baccalaureate programmes: <https://ibo.org/programmes/>

Repository sources

- components/EducationPathwayFields.tsx
- lib/educationTaxonomy.ts
- lib/lesson-request-intake.ts
- components/CalendlyBookingWizard.tsx
- app/free-assessment/FreeAssessmentForm.tsx
- app/tutor/actions.ts and app/free-assessment/actions.ts
- sql/051_lesson_request_intake.sql
- tests/education-taxonomy.test.mjs

Document owner

ScienceDojo Product Engineering. Update this guide whenever the intake schema, specification registry, plan output contract, or tutor approval workflow changes.